# Ahmad Abdal Qader

aabdalq@emory.edu | qader.dev | Atlanta, GA
linkedin.com/in/aaqader | orcid.org/0009-0001-3030-9864 | github.com/X4ndri

## Education

**Georgia Institute of Technology & Emory University** — *Atlanta, GA*
*Ph.D., Biomedical Engineering* — *Aug 2022 – Dec 2027 (expected)*
Computational Neural Engineering. Advised by Chethan Pandarinath and Ellen J. Hess

**University of Rochester** — *Rochester, NY*
*B.S., Biomedical Engineering — Signals and Systems* — *Aug 2018 – May 2022*
Cum Laude, General and Departmental Honors

## Publications

[J1] Y. H. Ali*, **A. Abdal Qader***, P. H. Bechefsky, B. A. Richards, A. Acosta, B. Lacayo, Z. M. Williams, L. R. Hochberg, N. AuYong, C. Pandarinath. Discrete spatial navigation as a complement to point-and-click in brain–computer interfaces. *Journal of Neural Engineering*, in revision. *equal contribution.

[J2] **A. Abdal Qader**, Y. Donsante, J. E. Markowitz, H. A. Jinnah, C. Pandarinath, E. J. Hess. Behavioral signature of trihexyphenidyl in the *TOR1A* (DYT1) knockin mouse model of dystonia. *Dystonia* 4:15034, 2025.

### Conference abstracts

[C1] **A. Abdal Qader**, Y. Donsante, J. E. Markowitz, H. A. Jinnah, C. Pandarinath, E. J. Hess. Behavioral signature of trihexyphenidyl in the *TOR1A* (DYT1) knockin mouse model of dystonia. *Interface Neuro*, Atlanta, GA, May 2025. Poster.

## Research Experience

**Emory University & Georgia Institute of Technology** — *Atlanta, GA*
*Graduate Research Assistant — Systems Neural Engineering Lab* — *Aug 2022 – Present*

**Sample-efficient neural handwriting decoding and cross-modal transfer**

- Collected a 256-channel intracortical attempted-handwriting dataset from a BrainGate2 participant with tetraplegia; trained and deployed sequence decoders reaching 6% offline and 14% online character error rate.
- Trained a cross-modal adapter into Meta's open-source surface EMG handwriting decoder, reusing large-scale non-invasive pretraining for intracortical decoding. Up to 4× more data-efficient than a tuned GRU trained from scratch, and generalized to characters absent from the neural training set.

**Real-time surface EMG acquisition and decoding in tetraplegia**

- Achieved closed-loop 1-D cursor control from residual, low-SNR surface EMG in a participant with tetraplegia and little functional hand movement, using an online decoding pipeline for a 95-channel wearable band.
- Integrated a wearable wristband recording sEMG, PPG, IMU and EDA, synchronized to simultaneous intracortical signals, toward a multimodal dataset relating neural activity to user state.

**Hybrid discrete–continuous BCI-driven on-screen navigation**

- Combined velocity decoding and gesture classification into a hybrid navigation paradigm with on-demand switching between fast cursor reaches and precise gesture-based selection.
- Trained gesture decoders on accumulated closed-loop data that generalized zero-shot to later sessions, staying accurate over weeks despite distribution shift in the input intracortical signals.

**Unsupervised modeling of behavior and neural activity in dystonia models**

- Quantified how an anti-dystonia drug reorganizes spontaneous mouse behavior by modeling 3D depth video as sequences of sub-second motifs with a hierarchical autoregressive HMM (MoSeq), uncovering consistent drug effects across genotypes and sex differences. Led data acquisition and analysis; first-author paper.
- Extracted single-neuron activity from low-SNR one-photon calcium imaging of the striatum with CNMF-E, including motion correction, parameter tuning and batch processing, plus a simulator for controlled testing.

**University of Rochester** *Rochester, NY*
*Research Assistant — Padmanabhan Lab* *May 2021 – May 2022*

- Implemented dynamic segmentation to extract spatially varying transient signals from fluorescence microscopy of morphing cells, with in-house acquisition and analysis tooling around it. Cultured human stem cell lines, ran viral transfections and imaging experiments.

*Research Assistant — Henry Lab* *Jun 2020 – Aug 2020*

- Built analog signal-conditioning circuits and 3D-printed parts to make reward delivery in a behavioral conditioning apparatus measurable and accurate; ran the experiments on avian models of hidden hearing loss.

## Conferences and Meetings

| | |
|---|---|
| Society for Neuroscience (SfN) | *Washington, DC / Nov 2026* |
| BrainGate Annual Summit | *Santa Cruz, CA / Aug 2026* |
| BrainGate Annual Summit | *Olympic Valley, CA / Aug 2025* |
| Interface Neuro Annual Meeting | *Atlanta, GA / May 2025* |
| BrainGate Annual Summit | *Atlanta, GA / Aug 2023* |
| Simons Foundation Workshop on Calcium and Voltage Imaging Analysis | *New York, NY / Feb 2023* |
| NSF Collaborative Research in Computational Neuroscience (CRCNS) | *Atlanta, GA / Oct 2022* |
| Simons–Emory International Consortium on Motor Control | *Atlanta, GA / Oct 2022* |

## Teaching and Mentoring

**Teaching Assistant** — Introduction to Biomechanics, Georgia Tech *Fall 2023, Spring 2024*

- Led the Problem Solving Studio section for 50 undergraduates working in groups of four to five, introducing open-ended problems and running solution reviews. Designed exam and quiz rubrics and supervised the undergraduate teaching assistants who graded.

**Undergraduate research mentorship** *2023 – Present*

- Mentored three undergraduate researchers on data acquisition, analysis pipelines and presentation of results.

## Awards and Fellowships

| | |
|---|---|
| **Computational Neural Engineering Training Program Scholar** (T32) | *Aug 2022* |
| **Cum Laude, General and Departmental Honors**, University of Rochester | *May 2022* |
| **Davis Scholar** — $80,000 undergraduate tuition award | *Aug 2018* |
| **International Baccalaureate Merit Scholarship** — $20,000 | *Aug 2018* |
| **Aurora 100Lives Gratitude Scholarship** — $70,000, United World College Dilijan | *May 2016* |

## Technical Skills

| | |
|---|---|
| **Concepts** | deep learning, time-series sequence modeling, transfer learning and domain adaptation, unsupervised learning, probabilistic graphical models, real-time signal processing |
| **Architectures** | RNNs, CNNs, transformers, state-space models, autoregressive HMMs |
| **Modalities** | intracortical signals, high-density sEMG, IMU, PPG, EDA; 3D depth video, calcium imaging |
| **Tools** | PyTorch, NumPy, SciPy, Weights & Biases, Redis, BRAND, DeepLabCut, CaImAn |
| **Languages** | Python, MATLAB, bash, LaTeX |
| **Hardware** | real-time data acquisition, multi-electrode arrays, circuit design, microcontrollers |

## Other Experience

**AS&E Information Technology Center**, University of Rochester *Aug 2018 – May 2022*
IT Lead and Training Coordinator (2019–2022), previously Operations and End-User Support Consultant. Supervised and trained a team of student IT workers and planned the department's transition to remote operations in 2020.